

数书九章

Mathematical Treatise in Nine Sections

Fascicle I – The Dayan Category

Bilingual Edition / 双语版

By **Qin Jiushao** (秦九韶)

c. 1202–1261 CE

Presented in the 7th year of Chunyou (淳祐七年), 1247 CE

Left side: Original Classical Chinese text

Right side: English translation with modern mathematical commentary

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数书九章序

数书九章者，南宋秦九韶之所著也。秦氏字道古，鲁郡人。生于宋季，博学多能，尤精算学。此书成于淳祐七年（公元1247年），凡九卷，每卷

每问有问、答、术、草四目，体例谨严。大衍之类，尤推精妙，盖中国剩余定理之滥觞也。

1.1 The Nine Categories / 九类名目

九类之目：

1. **大衍** – 求一之术
2. **天时** – 天文历法
3. **田域** – 田地测量
4. **测望** – 勾股重差
5. **赋役** – 租税劳役
6. **钱谷** – 钱粮贸易
7. **营建** – 建筑工事
8. **军旅** – 兵家之事
9. **市易** – 市场交易

Preface to the Shuxue Jiuzhang

The Shuxue Jiuzhang (Mathematical Treatise in Nine Chapters) is a significant work by the Song Dynasty mathematician Qin Jiushao (秦九韶, c. 1202–1261). Completed in 1247 CE during the Chunyou era (淳祐七年), it stands as one of the greatest achievements in the history of Chinese mathematics and indeed of world mathematics.

The work is organized into nine categories (类), each containing problems (问), for a total of eighty-one problems. Each problem follows a rigorous four-part structure: **Wen** (问) – the problem statement; **Da** (答) – the numerical answer; **Shu** (术) – the general method; **Cao** (草) – the detailed working.

The nine categories are:

1. **Dayan** (大衍) – “Great Derivation” : Indeterminate analysis and the Chinese Remainder Theorem
2. **Tianshi** (天时) – Astronomical phenomena and calendar calculations
3. **Tianyu** (田域) – Land measurement and area calculation
4. **Cewang** (测望) – Surveying and observation
5. **Fuyi** (赋役) – Taxation and labor service
6. **Qian’ gu** (钱谷) – Currency and grain commerce
7. **Yingjian** (营建) – Construction and engineering
8. **Junlv** (军旅) – Military affairs
9. **Shiyi** (市易) – Market transactions

2 Preface / 序

周教六艺，数实成之。学士大夫所从事尚矣。其用本太虚生一，而周流无穷。

Of the Six Arts taught by the Zhou, mathematics completes them. It has long been pursued by scholars and gentlemen. Its use originates from the One born of the Great Void, flowing endlessly. In the large, it can penetrate the divine and accord with nature; in the small, it can govern worldly affairs and classify all things. How could it be viewed as shallow or trivial?

若昔推策以迎日，定律而和气。髀距濬川，土圭度晷。天地之大，圉焉而不能外，况其间总悉者乎？爰自河图洛书，闡发幽秘，八卦九畴

From ancient times, reckoning rods were used to predict the sun's course, and laws were established to harmonize the qi. The gnomon measured the rivers, the jade tablet measured the shadows. The greatness of heaven and earth is encompassed by it and cannot escape, how much more the multitude of things within?

极而至于大衍、皇极之用，而人事之变无不该，鬼神之情莫能隐矣。圣人神之言而遗其粗，常人昧之由而莫之觉。要其归，则数与道非二

Reaching to the applications of the Dayan and the Supreme Ultimate, there is no human affair that it does not encompass, no spirit's nature that it cannot reveal. The sages found its spirit and left its roughness; ordinary people are blind to its path. In essence, number and the Way are not two different roots.

独大衍法不载九章，未有能推之者。历家演法颇用之，以为方程者，误也。

Only the Dayan method is not recorded in the Nine Chapters, and none have been able to develop it. Calendarists have used it extensively in their computational methods, but those who take it as a system of equations are mistaken.

九韶愚陋，不闲于艺。然早岁侍亲中都，因得访习于太史，又尝从隐君子受数学。时陈兵难，历岁遯塞，不自意至于矢石之间。更险离忧

I Jiushao am foolish and unrefined, not skilled in the arts. Yet in my youth I served my parents in the capital, and so had the opportunity to study with the Grand Astrologer; I also received instruction in mathematics from a gentleman recluse. When the time of military disaster came, I did not expect to survive between arrows and stones. Passing through danger and sorrow, ten years elapsed; my heart withered and my spirit fell. Truly I came to know that all things have number.

时淳祐七年九月，鲁郡秦九韶叙。

Written in the ninth month of the seventh year of Chunyou, by Qin Jiushao of Lu Commandery.

3 The Dayan General Method / 大衍总数术

3.1 Editor’s Commentary / 按语

按大衍术，以各分数之奇零求各分数之总数。大而天行，小而物数，皆可御之。	Editor’s note: The Dayan method takes the remainders (odd fragments) of each fractional divisor to find the total number. Whether for large celestial motions or small material quantities, it can be applied. The method has the following steps: finding the original numbers, finding the definitive numbers, finding the derivative numbers, finding the odd numbers, finding the multipliers, and finding the use numbers.
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3.2 The Four Types of Numbers / 四种问数

大衍总数术曰：置诸问数，类名有四。

- 1. 元数 – 谓尾位见单零者
- 2. 收数 – 谓尾位见分厘者
- 3. 通数 – 谓诸数各有分子分母者
- 4. 复数 – 谓尾位见十或百及千以上者

The Dayan General Method states: Place the given numbers; there are four types.

- 1. **Original numbers** – Numbers ending in single units
- 2. **Collected numbers** – Numbers ending in fractional parts
- 3. **Common numbers** – Numbers each having numerator and denominator
- 4. **Repeated numbers** – Numbers ending in tens, hundreds, or thousands

3.3 Finding the Definitive Numbers / 求定数

元数者：先以两两连环求等，约奇弗约偶。或约得五而彼有十，乃约偶弗约奇。或无数偶偶，约之可存一位见偶。

For original numbers: First, pairwise in chain sequence reduce the odd, not the even one. If reducing yields 5 while the other has 10, then reduce the even one, not the odd one. If all original numbers are even, after reduction one may keep one even number.

复数者：问数尾位见十以上者，以诸数求总等，存一位约众位，始得元数。两两连环求等，约奇弗约偶。复乘偶，或约偶弗约奇。

For repeated numbers: When the given numbers end in tens or above, find the GCDs; reduce the odd, not the even, and multiply back the even; or reduce the even, not the odd, and do not multiply back the odd.

3.4 Computing the Derivative Numbers / 求衍数

求定数勿使两位见偶，勿使见一太多。见一多则借用繁，不欲借则任得偶。

以定相乘为衍母，以各定约衍母，得各衍数。

次以各定母满去衍数，各余名曰奇数。

以大衍求一术，得各乘率。

以乘衍数，各得用数。

Finding definitive numbers: Do not let two positions be even; do not let there be too many ones. Too many ones makes borrowing complicated.

Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain each **derivative number**. Next, reduce each derivative number by its corresponding definitive number; the remainders are called the **odd numbers**.

Using the **Dayan Method for Finding Unity**, obtain each **multiplier**.

Multiplying each multiplier by its derivative number gives each **use number**.

3.5 The Dayan Method for Finding Unity / 大衍求一术

大衍求一术云：置奇右上，定居右下。立天元一于左上。先以右行上下两位，以少除多，所得商数，乃递互乘归左行。须使右上末后奇一归一。	The Dayan Method for Finding Unity says: Place the odd number at the upper right, the definitive number at the lower right. Place the celestial element unity (1) at the upper left. First, divide the larger by the smaller of the two numbers in the right column; the quotient obtained is then used to multiply alternately into the left column. Continue until the upper right finally becomes one. Then check what is obtained at the upper left—this is the multiplier.
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In modern mathematical terms: Given coprime integers a (the “odd number”) and m (the “definitive num-

ber”), find k such that:

$$k \cdot a \equiv 1 \pmod{m}$$

This is the **modular multiplicative inverse** of a modulo m , computed by the **extended Euclidean algorithm**.

4 Problem 1: Shigua Fawei / 蓍卦发微

蓍卦发微 – Divination by Yarrow Stalks	
问曰： 《易》曰：大衍之数五十，其用四十有九。又曰：分而为二以象两，挂一以象三，揲之以四以象四时。“由变而成，有变而成卦。欲知所衍	Question: This book, the Chongxing says: “Divide, of the Great Derivation is fifty; of these, forty-nine are used.” It also says: “Divide into two to symbolize the Two [heaven and earth]; hang one to symbolize the Three [heaven, earth, and man]; count off by fours to symbolize the Four Seasons.” Three changes produce a line; eighteen changes produce a hexagram. The problem asks to find the computational method of this derivation and the numbers involved. This problem applies the Dayan method to the classical yarrow-stalk divination of the I Ching.
答曰： 衍母十二，衍法三。 一元衍数二十四，二元衍数一十二，三元衍数八，四元衍数六。以上四位衍数，共五十一。 一揲用数一十二，二揲用数二十四，三揲用数四，四揲用数九。以上四位用数，共四十九。	Answer: The derivative mother is 12, the derivative divisor is 3. The first derivative number is 24, the second is 12, the third is 4, the fourth is 9. The sum of these four derivative numbers is 51. The first manipulation use-number is 12, the second is 24, the third is 4, the fourth is 9. The sum of these four use-numbers is 49.^a
术曰： 置诸元数，两两连环求等，约奇弗约偶。遍约毕，乃变元数皆曰定母，列在行各立天元一为子，列在行；以诸定母互乘左行之子，各得名曰衍	Method: Place the given original numbers, pairwise find the GCD, reducing the odd and not the even. When reduction is complete everywhere, the original numbers become the definitive numbers; place them in the right column. Place the celestial element unity (1) as the seed in the left column. Cross-multiply the definitive numbers with the seeds in the left column; each result is called a derivative number. Next, reduce each derivative number by its corresponding definitive number; each remainder is called the odd number. Using the Dayan Method for Finding Unity on the odd number and the definitive number, obtain each multiplier. Multiply each derivative number by its multiplier to obtain each use number.
次以各定母满去衍数，各余名曰奇数。以奇数与定母用大衍术求一，得各乘率以乘衍数，各得用数。	

诸定相乘为衍母 Dayan 定约衍母得衍数，各满定母去之，余为奇数。大衍求

the original calendar; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought accumulated years.

6 Problem 3: Tuiji Tugong / 推计土功

推计土功 – Estimating Earthworks	
问曰： 四县（甲、乙、丙、丁）共同筑堤，根据各县的物力分配筑堤任务。 - 甲县物力：138,600贯 - 乙县物力：146,300贯 - 丙县物力：192,500贯 - 丁县物力：184,800贯 筑堤标准：每770贯物力分配一名劳动力，每人每天筑堤60平方尺。 进度情况：甲县和乙县已完成任务，丙县剩余51丈，丁县剩余18丈。	Question: Four counties (A, B, C, D) jointly build a dike; the task is distributed according to each county’ s resources. - County A resources: 138,600 guan - County B resources: 146,300 guan - County C resources: 192,500 guan - County D resources: 184,800 guan Dike standard: for every 770 guan of resources, assign one laborer; each laborer builds 60 square feet of dike per day. Progress: Counties A and B have completed their tasks; County C has 51 zhang remaining; County D has 18 zhang remaining. This is a classic indeterminate problem solved by the Dayan method.
答曰： 堤的总长度为19里235步5尺。 各县所筑长度： - 甲县：1,260丈 - 乙县：1,768步5尺6寸 - 丙县：4里328步5尺6寸 - 丁县：与前三县相应	Answer: The total length of the dike is 19 li 235 bu 5 chi. Each county’ s contribution: - County A: 1,260 zhang - County B: 1,768 bu 5 chi 6 cun - County C: 4 li 328 bu 5 chi 6 cun - County D: corresponding to the first three
术曰： 以大衍求之。置各县物力，连环求等，约奇弗约偶，得定母。诸定相乘为衍母，以各定约衍母得衍数。各满定母去之，余为奇数。	Method: Solve by the Dayan Method. Place the resources of each county; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each county’ s remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

约奇非的偶，得定母，诸定相乘为行母，以定约行母得衍数。各满定母去

treasury A; subtracting the decrement each time gives each treasury's rate. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

8 Problem 5: Fentiao Tuiyuan / 分棗推原

分棗推原 – Grain Distribution	
问曰： 有兄弟三人，各将收获之米送往不同的地方： • 甲将米送往官场，余3斗2升 • 乙将米送往安吉乡，余7斗 • 丙将米送往平江，余3斗 官斛每石八斗三升，安吉斛每石一石一斗，平江斛每石一石三斗五升。欲求三人共收获米数以及各自分米数几何？	Question: Three brothers each send their harvested rice to different places: • A sends rice to the government office, remainder 3 dou 2 sheng • B sends rice to Anji town, remainder 7 dou • C sends rice to Pingjiang, remainder 3 dou The official measure holds 8 dou 3 sheng per shi; the Anji measure holds 1 shi 1 dou; the Pingjiang measure holds 1 shi 3 dou 5 sheng. Find: the total amount of rice harvested by the three brothers, and the amount each brother received.
答曰： 三人共收获米738石。 各分米数： • 甲：296石，余3斗2升 • 乙：223石，余7斗 • 丙：182石，余3斗	Answer: The three brothers harvested 738 shi of rice in total. Each brother's share: • A: 296 shi, remainder 3 dou 2 sheng • B: 223 shi, remainder 7 dou • C: 182 shi, remainder 3 dou
术曰： 以大衍求之。置各斛率（单位：升）：官斛83升，安吉斛110升，平江斛135升。每斗以十，每升以十，得各斛率。置各斛率，以各定约	Method: Solve by the Dayan method. Each measure has a rate (in sheng): the official measure 83 sheng, the Anji measure 110 sheng, the Pingjiang measure 135 sheng. Pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers.

9 Problem 6: Chengxing Jidi / 程行计地

程行计地 – Journey Distance Calculation

问曰： 军师派遣三名急足（甲、乙、丙）分别前往都下报捷。 • 甲每天行300里 • 乙每天行240里 • 丙每天行180里 甲提前数日申未到，乙后数日未正到，丙今日辰未到。欲知从军前到都下的距离以及三人各自行走的天数几何？	Question: A military commander sends three express messengers (A, B, C) to the capital to report victory. • A travels 300 li per day • B travels 240 li per day • C travels 180 li per day A arrives early (before shen hour); B arrives late (at wei hour); C arrives today (at chen hour). Find the distance from the army's position to the capital, and the number of days each traveled.
答曰： 从军前到都下的距离为3,300里。 • 甲行11天 • 乙行13天4时 • 丙行18天2时	Answer: The distance from the army to the capital is 3,300 li. • A traveled 11 days • B traveled 13 days 4 hours • C traveled 18 days 2 hours
术曰： 以大衍求之。置三人日行里数，连环求等，约奇弗约偶，得定母。诸定母乘之行母，大衍定约得衍数，各满定母去之，余为奇数，以大衍求之。置三人日行里数，连环求等，约奇弗约偶，得定母。诸定母乘之行母，大衍定约得衍数，各满定母去之，余为奇数，以大衍求之。	Method: Pairwise find GCDs of travel distances; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought distance.

10 Problem 7: Chengxiang Xiangji / 程行相及

程行相及 – Catching Up on a Journey	
问曰： 甲、乙、丙三人行程相关。甲先行若干日，乙后行，丙又后行。各以不同速度行进，求某处，三人相遇之时间及各地距离。	Question: Three travelers, A, B, and C, start on different journeys. A departs first, B departs later, and C departs still later. Each travels at a different speed. Find: the time when all three meet at a certain place, and the various distances. This problem involves multiple travelers with different speeds and starting times, requiring the Dayan method to solve a system of congruences.
术曰： 以大衍求之。置各人行程速度及时间差，连环求等，约奇弗约偶，得定母。诸定母相乘，以各定约行母得衍数。各定母去之，余为奇数。	Method: Solve by the Dayan method. Place each person's speed and time difference; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder is the sought answer.

11 Problem 8: Jichi Xunyuan / 积尺寻源

积尺寻源 – Finding Dimensions from Volume

问曰： 问欲砌基一段，见管大小方砖六门城砖四色。令匠取便，或平或侧，只用一色砖砌，须要适足。匠以破量地计料，并用 • 大方：广多六寸，深少六寸 • 小方：广多二寸，深少三寸 • 城砖：长广多三寸，深少一寸 • 六门砖：长广多三寸，深多一寸 皆不合匝，未免修破砖料裨补。其四色砖尺寸： • 大方：方一尺三寸 • 小方：方一尺一寸 • 城砖：长一尺二寸，阔六寸，厚二寸五分 • 六门砖：长一尺，阔五寸，厚二寸 欲知基深广几何？ 答曰： 深三丈七尺一寸，广一丈二尺三寸。 术曰： 以大衍求之。置四色砖尺寸及所余尺寸，连环求等，约奇弗约偶，得定	Question: One mason, to build a foundation, has available four types of bricks: large and small square bricks, six-portal bricks, and city-wall bricks—four types in all. The mason is instructed to use whichever is convenient, laying them flat or on edge, using only one type of brick, requiring them to fit exactly. The mason measured the ground and estimated materials, reporting: • Large square: width excess 6 cun, depth deficit 6 cun • Small square: width excess 2 cun, depth deficit 3 cun • City brick: length and width excess 3 cun, depth deficit 1 cun • Six-portal: length and width excess 3 cun, depth excess 1 cun All fail to fit exactly, so broken bricks must be used for patching. The four types of brick have dimensions: • Large square: 1 chi 3 cun square • Small square: 1 chi 1 cun square • City brick: 1 chi 2 cun long, 6 cun wide, 2 cun 5 fen thick • Six-portal: 1 chi long, 5 cun wide, 2 cun thick Find: the depth and width of the foundation. This is a classic problem of finding a common multiple with given remainders. Answer: Depth: 3 zhang 7 chi 1 cun; width: 1 zhang 2 chi 3 cun. Method: Solve by the Dayan Method. Place the four types of brick dimensions and the excess/deficit dimensions; pairwise in chain sequence find GCDs, reducing the odd and not the even, obtaining the definitive numbers. Multiply the definitive numbers together to obtain the derivative mother. Reduce the derivative mother by each definitive number to obtain the derivative numbers. Reduce each by its definitive number; the remainders are the odd numbers. Apply the Dayan Method for Finding Unity to obtain the multipliers. Multiply the derivative numbers by the multipliers to obtain the use numbers. Next, place each remainder, multiply by the use numbers, and sum to obtain the total number. Reduce by the derivative mother; the remainder gives the depth and width of the foundation.
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12 Problem 9: Yumi Tuishu / 余米推数

余米推数 – Calculating Rice Remainders

问曰： 有米若干，以不同容量的容器量之，各余若干。欲求米之总数。	Question: There is a quantity of rice; when measured with containers of different capacities, there are different remainders each time. Find the total amount of rice. This is the classic “unknown quantity” problem. In modern terms: find N such that $N \equiv r_i \pmod{m_i}$ for given moduli m_i and remainders r_i.
术曰： 以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。	Method: 以大衍求之。置各容器容量为问数，连环求等，约奇弗约偶，得定母。此即大衍求一术。以各定约母得衍数，各满定母去之，余为奇数。大衍求一术，即今之中国剩余定理。以各奇数乘衍数，相加，得用数。以用数乘定母，得总数。以总数减衍数，得米之总数。

This problem is structurally identical to the famous “problem of the unknown quantity” from the Sunzi Suanjing (孙子算经), which asks: “We have things of which we do not know the number; if we count them by threes, the remainder is 2; if by fives, the remainder is 3; if by sevens, the remainder is 2. How many things are there?” Qin Jiushao’s general method solves all problems of this type systematically.

13 Mathematical Analysis / 数理分析

13.1 The Chinese Remainder Theorem / 中国剩余定理

The Dayan method solves systems of simultaneous linear congruences. Given a set of pairwise coprime moduli $m_1, m_2, \dots, m_n$ and remainders $r_1, r_2, \dots, r_n$, find N such that:

$$N \equiv r_1 \pmod{m_1}$$
$$N \equiv r_2 \pmod{m_2}$$
$$\vdots$$
$$N \equiv r_n \pmod{m_n}$$

13.2 Qin Jiushao’s Algorithm / 秦九韶算法

- 求定数 – 连环求等，约奇弗约偶
- 求衍母 – 诸定相乘
- 求衍数 – 以定约衍母
- 求奇数 – 各满定母去之
- 求乘率 – 大衍求一术
- 求用数 – 乘率乘衍数
- 求总数 – 余数乘用数并之

- Step 1: Definitive Numbers. Given the original numbers (元数), reduce them pairwise using the GCD, keeping them coprime. The rule “reduce the odd, not the even” ensures unique factorization.
- Step 2: Derivative Mother. Compute $M = m_1 \times m_2 \times \dots \times m_n$.
- Step 3: Derivative Numbers. Compute $M_i = M/m_i$ for each i .
- Step 4: Odd Numbers. Compute $g_i = M_i \bmod m_i$.
- Step 5: Finding Unity. For each i , find k_i such that $k_i \cdot g_i \equiv 1 \pmod{m_i}$. This is the modular multiplicative inverse, computed by the extended Euclidean algorithm.
- Step 6: Use Numbers. Compute $u_i = k_i \cdot M_i$.
- Step 7: Final Computation. The solution is: $N = \sum_{i=1}^n r_i \cdot u_i \pmod{M}$

13.3 Historical Significance / 历史意义

秦氏大衍术，超前欧洲数百年。其求一之术，即欧几里得之扩展算法也。后世郭守敬用之于天文学，李冶用于几何，在西传欧洲，遂成所谓“中国剩余定理”。高斯于《算术探究》（1801年）首次在欧洲完整论述此定理，称之为“求一术”（以备已知数除之，各余同知数）问题。今此定理为数论之基本算法。The method was later applied by Guo Shoujing (郭守敬) to astronomy, by Li Ye (李冶) to geometry, and was transmitted to Europe where it became known as the “Chinese Remainder Theorem.” Gauss gave the first complete treatment in Europe in his Disquisitiones Arithmeticae (1801), calling it the “problem of finding a number that, when divided by given numbers, leaves given remainders.” The theorem is now recognized as one of the fundamental results in number theory.

About This Edition / 关于本版

This bilingual edition presents Fascicle I of Qin Jiushao’s *Shuxue Jiuzhang* (Mathematical Treatise in Nine Sections), with the original classical Chinese text in the **left column** and an English translation with modern mathematical commentary in the **right column**.

Technical Notes / 技术说明

- Typeset with XeTeX using parallel `minipage` environments
- Font: Noto Sans CJK SC (supports both Chinese and Latin characters)
- The traditional structure of 问 (question), 答 (answer), 术 (method), and 草 (working) has been preserved
- Explanatory notes explain technical terminology and historical context

Explanatory notes / 翻译说明

- “Dayan” (大衍) — translated as “Great Derivation,” preserving the I Ching reference
- “Qiuyi” (求一) — translated as “Finding Unity” (finding the modular inverse)
- “Yanshu” (衍数) — translated as “derivative numbers”
- “Yanmu” (衍母) — translated as “derivative mother”
- “Chengshu” (乘率) — translated as “multiplier”
- “Yongshu” (用数) — translated as “use number”

Further Reading / 参考文献

- Libbrecht, U. *Chinese Mathematics in the Thirteenth Century*. MIT Press, 1973.
- Qian, Baocong. *Zhongguo Shuxue Shi* (History of Chinese Mathematics). Science Press, 1964.
- Martzloff, J.-C. *A History of Chinese Mathematics*. Springer, 1997.